Chapter 4 · Graphs and transformations

starter

1(a) Factorise $x^2 + 6x + 5$.

1(a)

1(b) Factorise $x^2 - 4x + 3$.

1(b)

2(a) Sketch $y = (x + 2)(x - 3)$.

2(a)

2(b) Sketch $y = x^2 - 6x - 7$.

2(b)

3 Complete a table and draw $y = x^3 + x - 2$ for $x = -2, -1.5, -1, -0.5, 0, 0.5, 1, 1.5, 2$.

3

3 sketch

Chapter Overview 3

1. Sketching graphs

Cubic, quartic and reciprocal functions: roots, intercepts, repeated roots and end behaviour.

2. Points of intersection

Connect graphical intersections with simultaneous equations.

3. Transformations

Translate, stretch and reflect graphs. Track points, intercepts and asymptotes.

Polynomial Graphs 4

A polynomial has nonnegative integer powers of x:

$a + bx + cx^2 + dx^3 + ex^4 + \cdots$

Its **degree** is the highest power with a nonzero coefficient.

Degree 0

4

Degree 0

Degree 1

$2x - 1$

Degree 1

Degree 2

$x^2 + 3$

Degree 2

Degree 3

x^3

Degree 3

Degree 4

x^4

Degree 4

Degree 5

x^5

Degree 5

Turning points How does the degree restrict the shape?

Turning points

End behaviour What determines the direction of the tails?

End behaviour

An upward quartic need not have three turns. Do not infer the number of turns just from its degree.

For $y = ax^n + \cdots$, with $a \neq 0$, describe the left and right tails.

Degree 2, a>0

$y = ax^2 + \cdots$

Degree 2, a>0

Degree 2, a<0

$y = ax^2 + \cdots$

Degree 2, a<0

Degree 3, a>0

$y = ax^3 + \cdots$

Degree 3, a>0

Degree 3, a<0

$y = ax^3 + \cdots$

Degree 3, a<0

Degree 4, a>0

$y = ax^4 + \cdots$

Degree 4, a>0

Degree 4, a<0

$y = ax^4 + \cdots$

Degree 4, a<0

Degree 5, a>0

$y = ax^5 + \cdots$

Degree 5, a>0

Degree 5, a<0

$y = ax^5 + \cdots$

Degree 5, a<0

1 Sketch

$y = (x-2)(1-x)(1+x)$

1 features

1 sketch

2 Sketch

$y = x^2(x-1)$

2 features

2 sketch

1 Sketch

$y = (2-x)(x+1)^2$

1 features

1 sketch

2 Sketch

$y = (x-4)^3$

2 features

2 sketch

Cubics with Limited Roots 9

Sketch
$y = (x+1)(x^2+x+1)$

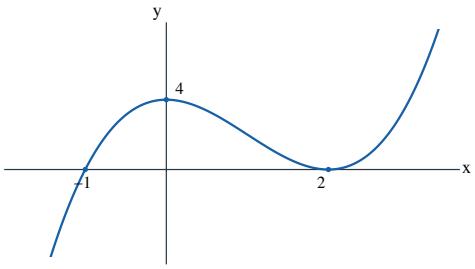
Sketch features

Sketch sketch

Later, differentiation confirms that there are no turning points:
$y\prime = 3x^2 + 4x + 2 = 3(x+\frac{2}{3})^2 + \frac{2}{3} > 0.$

Finding the equation yourself 10

A monic cubic crosses at $(-1, 0)$, touches at $(2, 0)$, and has a maximum at $(0, 4)$. Find a, b and c in $y = x^3 + ax^2 + bx + c$.



Answer

1. Sketch $y = x(x-3)^2$.

1

2. Sketch $y = -(x+2)^3$.

2

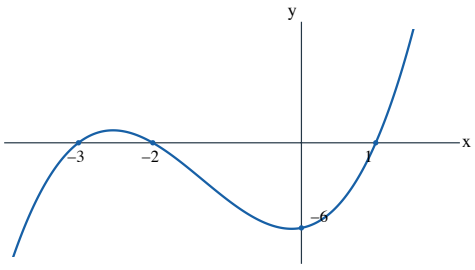
3. A positive cubic crosses at -2 and touches at 3. Give one suitable equation.

3

Extension. Sketch $y = 2x^2(x-1)(x+1)^3$.

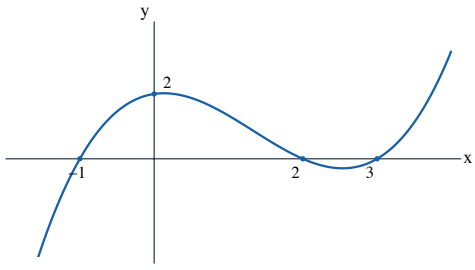
Extension

5. A monic cubic crosses the x-axis at -3, -2 and 1. (a) Find b, c and d in $y = x^3 + bx^2 + cx + d$. (b) Find its y-intercept.



5

6. A cubic crosses at -1, 2 and 3 and has y-intercept 2. Find a, b, c, d in $y = ax^3 + bx^2 + cx + d$.



6

$f(x) = (x - 10)(x^2 - 2x) + 12x$

7(a) Express f(x) in the form $x(ax^2 + bx + c)$.

7(a)

7(b) Factorise completely.

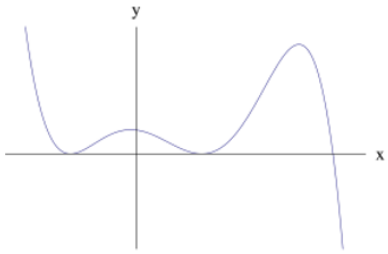
7(b)

7(c). Sketch the curve, marking all intercepts.

7(c)

1. MAT 2012 1E

Which equation could give this sketch?



A : $(3 - x)^2(3 + x)^2(1 - x)$

B : $-x^2(x - 9)(x^2 - 3)$

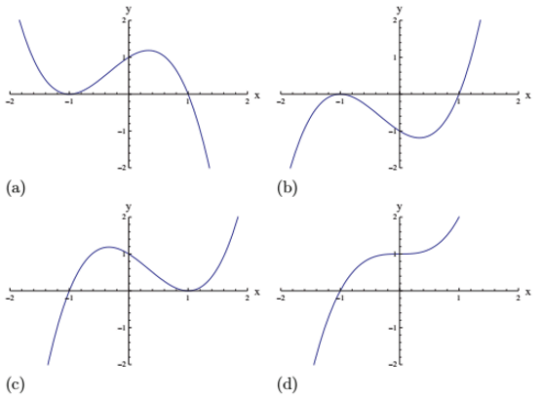
C : $(x - 6)(x - 2)^2(x + 2)^2$

D : $(x^2 - 1)^2(3 - x)$

1

2. MAT 2011 1A

Which sketch represents $y = x^3 - x^2 - x + 1$?



2

Recap 15

For distinct a, b, c, consider $y = (x-a)(x-b)^2(x-c)^3$. What happens at each root?

a A simple root.

a

b A double root.

b

c A triple root.

c

Quartics 16

1 Sketch

$y = x(x+1)(x-2)(x-3)$

1 features

1 sketch

2 Sketch

$y = (x-2)^2(x+1)(3-x)$

2 features

2 sketch

Quartics

17

1 Sketch

$y = (x+1)(x-1)^3$

1 features

1 sketch

2 Sketch

$y = (x-2)^4$

2 features

2 sketch

Test Your Understanding

18

1 Sketch

$y = x^2(x+1)(x-1)$

1 features

1 sketch

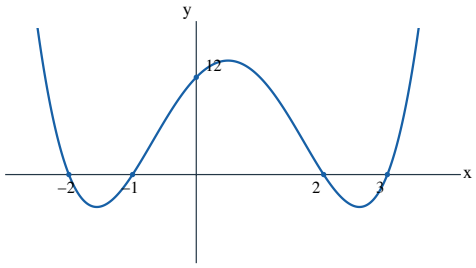
2 Sketch

$y = -(x+1)(x-3)^3$

2 features

2 sketch

3. The monic quartic $y = x^4 + bx^3 + cx^2 + dx + e$ has roots -2, -1, 2, 3. Find its y-intercept P and b, c, d, e.



3

4 Sketch
$y = (x + 5)(x - 4)(x^2 + 5x + 14)$

4 features

4 sketch

Exercise 4B Extension 20

STEP I 2012 Q2

(a) Sketch $y = x^4 - 6x^2 + 9$.

a

a features

(b) For which b does $x^4 - 6x^2 + b = 0$ have each number of distinct real roots?

b

GCSE RECAP: Reciprocal Graphs

1 Sketch

$y = \frac{1}{x}$

1 features

1 sketch

2 Sketch

$y = -\frac{3}{x}$

2 features

2 sketch

Both graphs have asymptotes $x = 0$ and $y = 0$. An asymptote is a line approached by a curve. These reciprocal graphs do not meet their asymptotes.

Reciprocal Graphs

1 Sketch

$y = \frac{3}{x^2}$

1 features

1 sketch

2 Sketch

$y = -\frac{4}{x^2}$

2 features

2 sketch

For both graphs the asymptotes are $x = 0$ and $y = 0$.

On the same axes sketch $y = 1/x$ and $y = 3/x$.

Sketch

Compare

Exercise 4C – Pure Y1 page 67 24

1. Sketch each pair on a separate set of axes. Blue is the first curve; brown is the second.

$y = 2/x, \quad y = 4/x$

a

$y = 2/x, \quad y = -2/x$

b

$y = -4/x, \quad y = -2/x$

c

$y = 3/x, \quad y = 8/x$

d

$y = -3/x, \quad y = -8/x$

e

2. Sketch each pair. Blue is the first curve; brown is the second. All have asymptotes x=0 and y=0.

$y = 2/x^2, \quad y = 5/x^2$ $y = 3/x^2, \quad y = -3/x^2$ $y = -2/x^2, \quad y = -6/x^2$

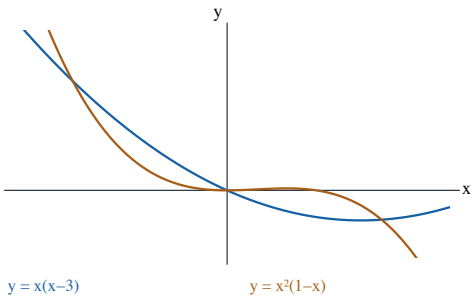
a

b

c

Points of Intersection 26

Find the intersections of $y = x(x - 3)$ and $y = x^2(1 - x)$.



Intersections

Further example involving unknown constants 27

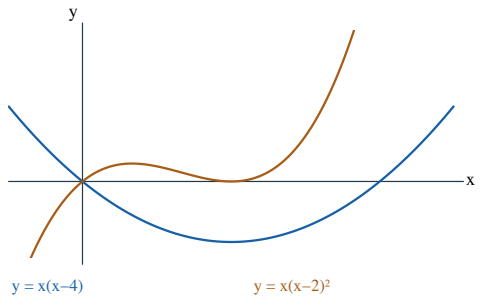
Sketch $y = x^2(3x - a)$ and $y = b/x$, where a and b are positive. How many intersections are there?

Sketch

Count

Test Your Understanding 28

Sketch $y = x(x - 4)$ and $y = x(x - 2)^2$. Find all intersections.



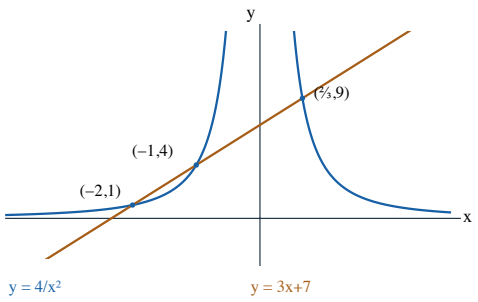
Answer

Sketch $y = x^2(x + a)$ and $y = b/x$, where a,b>0. State the number of intersections and form an equation for their x-coordinates.

Sketch

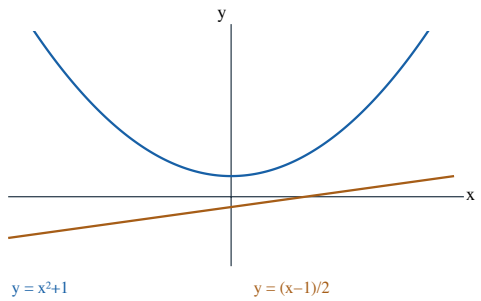
Answer

Find the intersections of $y = 4/x^2$ and $y = 3x + 7$.



Answer

1(a) Show that $y = x^2 + 1$ and $y = (x - 1)/2$ never meet.



a

1(b) For which a do $y = x^2 + a$ and $y = (x - 1)/2$ meet at two points?

1(b)

Sketch $y = x^2(x - 1)(x + 1)$ and $y = x^3/3 + 1$ on the same axes. How many real solutions does $x^2(x - 1)(x + 1) = x^3/3 + 1$ have?

Sketch

Answer

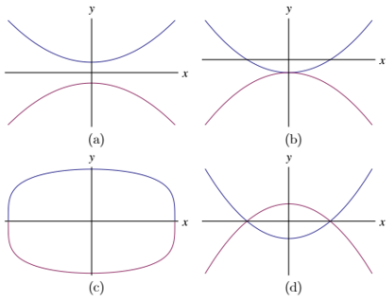
1. MAT 2005 1B

$(x^2+1)^{10} = 2x - x^2 - 2$

Does it have (A) solution 2, (B) no real solutions, (C) an odd number of real solutions, or (D) twenty real solutions?

1. MAT 2005 1B

2. MAT 2013 1D Which sketch represents $x^4 - y^2 = 2y + 1$?



2. MAT 2013 1D

3. MAT 2010 1A For which k does $y = kx$ intersect $y = (x-1)^2$?

3. MAT 2010 1A

For $f(x) = x^2$, compare $y = f(x)$ with $y = f(x+2)$.

Graphs

Transformation

Changes **inside** f affect x-coordinates inversely. Changes **outside** f act directly on y-coordinates.

1

$y = f(x - 3)$

1

2

$y = f(x) + 4$

2

3

$y = f(5x)$

3

4

$y = 2f(x)$

4

Sketching transformed graphs 36

1 Sketch

$y = x^2 + 3$

1 features

1 sketch

2 Sketch

$y = \dfrac{2}{x + 1}$

2 features

2 sketch

More Examples 37

1. Sketch $f(x) = x(x+2)$ and $y = f(x-a)$, a>2.

1

1 sketch

2. Sketch $f(x) = x^2(x-4)$ and $y = f(2x)$.

2

2 sketch

Reflections of Graphs 38

If $f(x) = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$.

Sketch

Answer

Test Your Understanding 39

1. Sketch $f(x) = (x+1)(x-2)$ and $y = f(x/3)$.

1

1 sketch

2 Sketch

$$y = \frac{2}{x} + 1$$

2 features

2 sketch

Exercise 4E – Pure Y1 page 74-75 (translations) 40

7. P(4,−1) lies on $y = f(x)$. Find its image.

7(a)

$y = f(x-2)$

7(a)

7(b)

$y = f(x) + 3$

7(b)

8 Translate $y = 1/x$ so its asymptotes are x=4 and y=0. Give its new equation.

8

11(a) Sketch $y = x^3 + 4x^2 + 4x$.

a

a sketch

11(b) Find both a for which $(-1,0)$ lies on $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$.

11(b)

12(a) Sketch $y = x(x + 1)(x + 3)^2$.

a

a sketch

12(b) Find all b for which $(2,0)$ lies on $y = (x + b)(x + b + 1)(x + b + 3)^2$.

12(b)

7. P$(2,-3)$ lies on $y = f(x)$. Find its image.

7(a)
$y = f(2x)$

7(a)

7(b)
$y = 4f(x)$

7(b)

8 Q$(-2,8)$ lies on $y = f(x)$. Find its image on $y = f(x/2)$.

8

9(a) Sketch $y = (x-2)(x-3)^2$.

a

9(b) The graph $y = (ax-2)(ax-3)^2$ passes through $(1,0)$. Find both a.

9(b)

Effect of transformation on specific points 45

Track these three points, in order: $(4,3),\ (1,0),\ (6,-4)$.

1

$y = f(x+1)$

1

2

$y = f(2x)$

2

3

$y = 3f(x)$

3

4

$y = f(x) - 1$

4

5

$y = f(x/4)$

5

6

$y = f(-x)$

6

7

$y = -f(x)$

7

Test Your Understanding 46

For $f(x) = x^2(9-2x)$, the minimum is (0,0) and maximum is (3,27).

a Find the other x-intercept A.

a

b(i) Sketch $y = f(x+3)$. Mark extrema and intercepts.

b(i)

b(i) sketch

b(ii) Sketch $y = f(3x)$. Mark extrema and intercepts.

b(ii)

b(ii) sketch

c $y = f(x) + k$ has maximum (3,10). Find k.

c

5. P(2,1) lies on $y = f(x)$.

a On $y = f(ax)$, P maps to Q(4,1). Find a.

a

b(i) Find the image on $y = f(x - 4)$.

b(i)

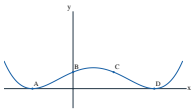
b(ii) Find the image on $y = 3f(x)$.

b(ii)

b(iii) Find the image on $y = f(x)/2 - 4$.

b(iii)

6. A curve $y = f(x)$ has points A(−1,0), B(0,2), C(1,2), D(2,0). Sketch each transformation and give the images of A, B, C, D.



$y + 2 = f(x)$

a

a sketch

$\frac{1}{2}y = f(x)$

b

b sketch

$y - 3 = f(x)$

c

c sketch

$3y = f(x)$

d

d sketch

$2y - 1 = f(x)$

e

e sketch